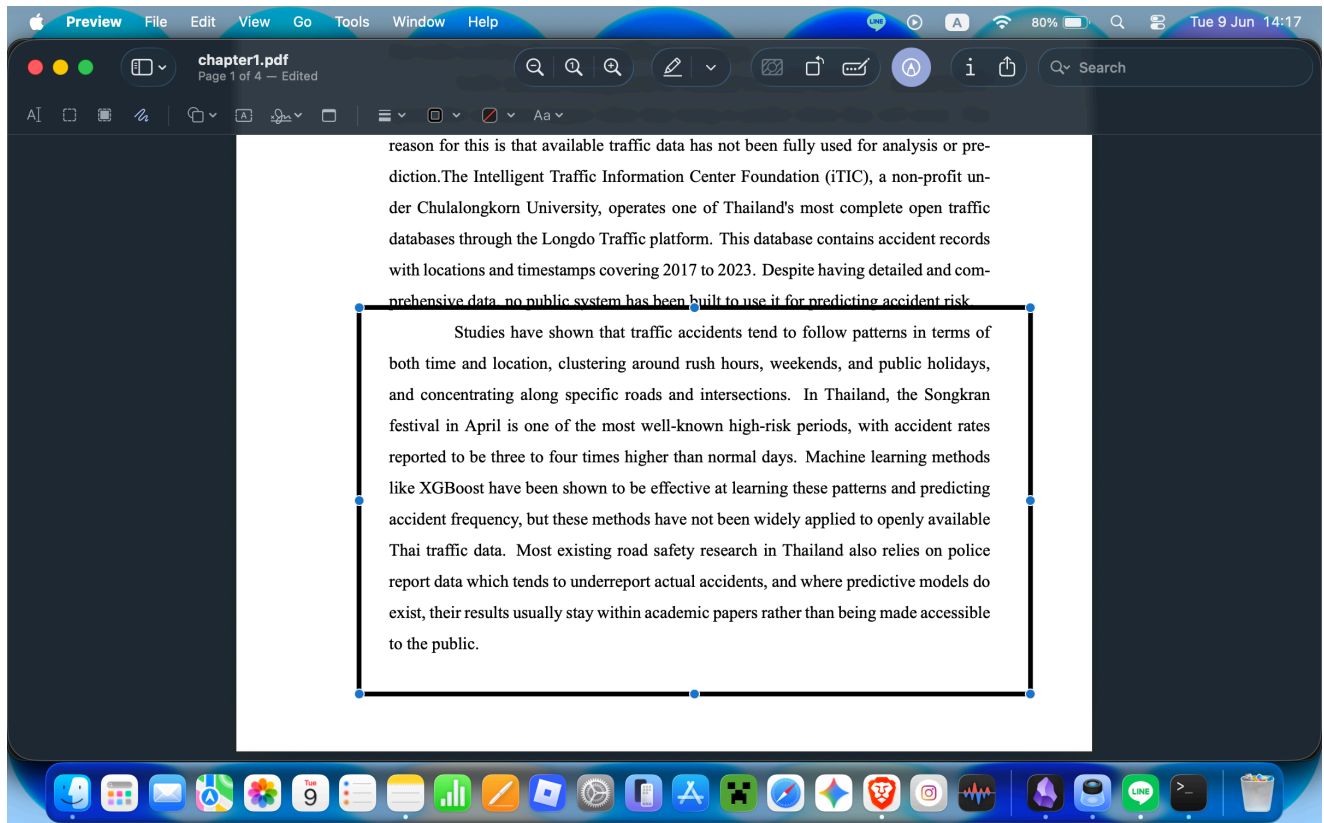


feedback01



This part should be moved to literature review

- From this part, the XGBoost model that is mentioned, there needs to be some kind of citation or reference to an academic paper, since although not mentioned, Ajarn feels like the source of information comes from a certain paper, so she wants to see that paper, and possible the reference of it as well (does not explicitly say that there needs to be citation)

reported to be three to four times higher than normal days. Machine learning methods like XGBoost have been shown to be effective at learning these patterns and predicting accident frequency, but these methods have not been widely applied to openly available Thai traffic data. Most existing road safety research in Thailand also relies on police

This project develops AcciSight, a data science system that uses historical accident records from iTIC to analyze and predict traffic accident risk in Bangkok. We explore patterns in the data across time and location, then train a machine learning model to forecast accident frequency across different zones in the city. The results are presented through an interactive web dashboard featuring a heatmap that shows both historical accident density and predicted risk areas, making it accessible for anyone to understand where and when accidents are most likely to occur.

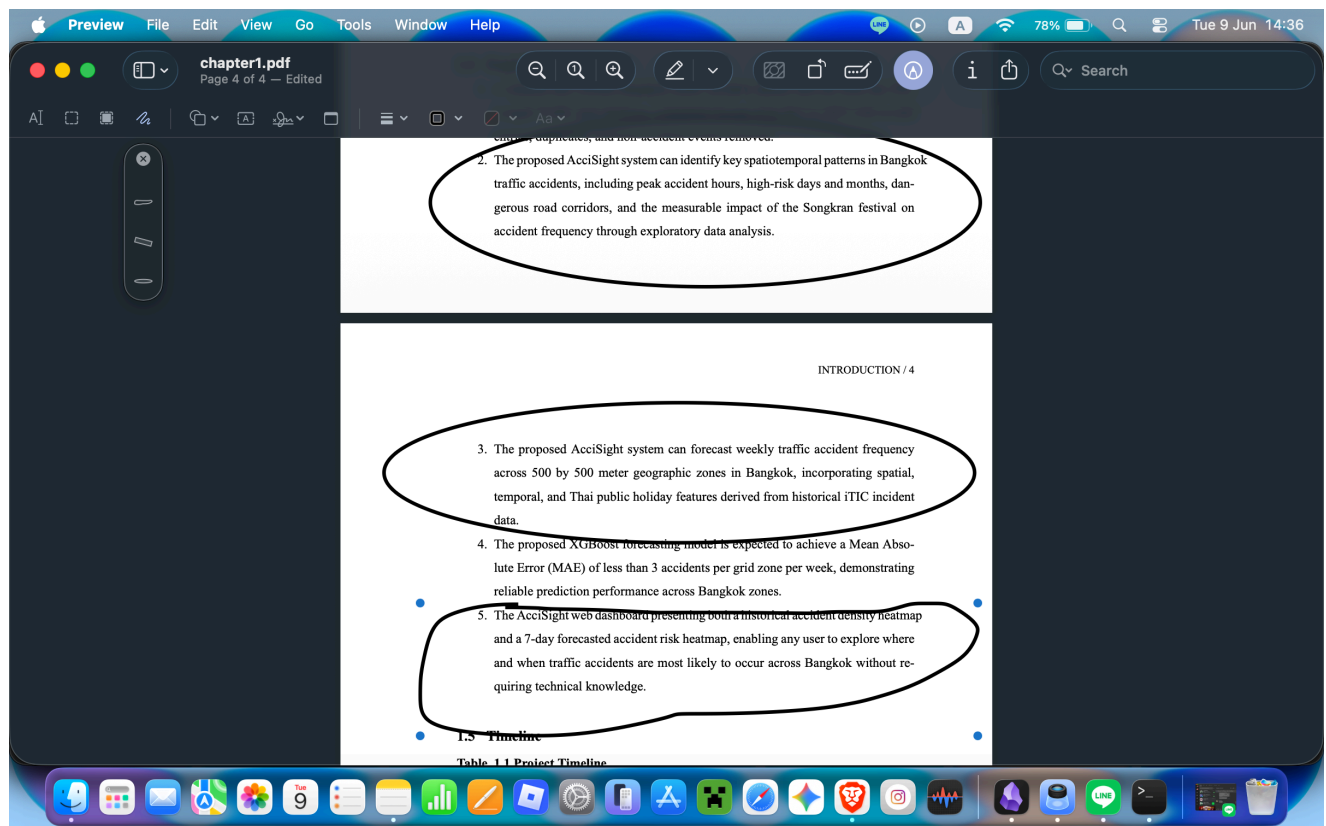
1.2 Objective

1. To develop a data cleaning and processing pipeline that extracts verified traffic accident records from the iTIC Longdo Traffic open dataset spanning 2017 to 2023, filtering out invalid coordinates, duplicate entries, and non-accident event types.
 2. To identify spatiotemporal patterns in Bangkok traffic accident occurrence through exploratory data analysis, examining accident frequency by hour of day, day of week, month of year, and Thai public holiday periods including the Songkran festival.
 3. To develop a grid-based machine learning forecasting model using XGBoost that predicts weekly traffic accident frequency across 500 by 500 meter geographic zones in Bangkok based on historical spatial, temporal, and holiday features.
 4. To improve forecasting accuracy by incorporating Thai public holiday features
- She asks whether we would have both accident frequency forecast and risk area indicator/forecast as well. I answered yes, she seems fine with it (she was confused of what exactly we are trying to predict, risk or frequency, I said that we predict risk, which would help us predict frequency, basically, predict two things.)
 - The factors which determine "risk" also needs to be further discussed.

The XGBoost machine learning model

- Because the chapter 1 only mentions XGBoost as our primary machine learning model to help processes the data, she said that we don't need to specify the model just yet, no need to commit to only XGBoost since we can test our data with different models as well, and see which one would work best, in terms of performance and UI compatibility (Accisight).
- We can use XGBoost paper (paper which implements the model) as the baseline, meaning we can use the same method as the paper, repeating basically, but we need to have something more than the paper, other than new UI and new dataset. Ajarn wants our project to have some contributions.
 - **My personal opinion** is that we don't need to specify machine learning model just yet, but during the testing period, we just use multiple models and test and documents our findings. With the last product as the documented comparison between models (our black book) and the UI which shows accidents forecast

She raises the question of whether Accisight is a web app, I answered yes.



These can be combined

We don't have to mention our method at the moment, I cannot remember which part she is referring to, but I guess it is among the same part as the part that needs combination, I think it is in the 4th.

data.

4. The proposed XGBoost forecasting model is expected to achieve a Mean Absolute Error (MAE) of less than 3 accidents per grid zone per week, demonstrating reliable prediction performance across Bangkok zones.
5. The AcciSight web dashboard presenting both a historical accident density heatmap and a 7-day forecasted accident risk heatmap, enabling any user to explore where and when traffic accidents are most likely to occur across Bangkok without requiring technical knowledge.

- "propose XGBoost" seems like we are implying new invention (or inventing XGBoost itself) the wording should be that we do
- "Comparison of the machine learning model for forecasting accident mix" and then continue, the rest seems ok.
- Ajarn raises a question again of **why** the metric to measure AI's performance has to be **Mean Absolute Error (MEA)**. Why that one specifically, and she mentioned that there can be more than 1 metrics.

She mentions the other machine learning model we can utilize, the new one, like ADABoost.

Ajarn mentions that it is crucial for us to document data cleaning process as I have mentioned that Darren and Kj have converted the files from .xml to csv and initialized the cleaning process already, the cleaning process would be mentioned in the thesis black book.